In-Class Practice: Oral Absorption

Module 5 - Single Oral Administration · In-class activity · 21 Sep 2026

Name _____ Date _____

Problem 1 A drug given orally has a half-life of absorption of 90 minutes. What is its absorption rate constant?

Problem 1 A 500-mg dose of a drug that is 88% bioavailable is given by mouth. Its half-life of absorption is 90 minutes and its half-life of elimination is 5 hours, and its apparent volume of distribution is 20 L. When does the peak concentration occur?

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Problem 1 For that same 500-mg oral dose, if the apparent volume of distribution were 40 L rather than 20 L, what peak plasma concentration would be expected?

Problem 2 A drug given as a 750-mg oral dose gives plasma concentrations described by $C_p = 23.2(e^{-0.182t} - e^{-0.872t})$, with t in hours. What is its elimination half-life?

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Problem 2 A drug given as a 750-mg oral dose gives plasma concentrations described by $C_p = 23.2(e^{-0.182t} - e^{-0.872t})$, with C_p in mg/L and t in hours. Calculate C_{max}.

Problem 2 A 750-mg oral dose of a drug that is 84% bioavailable gives plasma concentrations described by $C_p = 23.2(e^{-0.182t} - e^{-0.872t})$, with concentration in mg/L and time in hours. What is the apparent volume of distribution?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Problem 1 Worked by the drill

Answer: 0.462 hr^{-1}

WHAT THE STEM GIVES

$t_{½a}$	90 minutes	becomes 1.5 hr, the denominator of $\frac{0.693}{t_{½a}}$

THE WORKING, ONE LINE AT A TIME

UNIT

$$90\ \text{min} \times \frac{1\ \text{hr}}{60\ \text{min}} = 1.5\ \text{hr}$$

The rest of the problem is worked in hours, so the absorption half-life has to be expressed in hours before the rate constant is taken from it. Minutes multiplied by hours per minute cancels minutes and leaves hours. Ninety minutes is one and a half hours.

SETUP

$$k_a = \frac{0.693}{t_{½a}}$$

Absorption is first order, so its rate constant is 0.693 divided by its half-life. The number 1.5 is a half-life, not a rate constant, and using it directly in the oral equations is the error that this conversion exists to prevent.

ALGEBRA

$$k_a = \frac{0.693}{1.5\ \text{hr}} = 0.462\ \text{hr}^{-1}$$

Dividing 0.693 by hours leaves reciprocal hours. Carrying the 90 minutes into this line instead of the 1.5 hours would give 0.0077, a constant per minute rather than per hour. The rate constant is 0.693 times the reciprocal of the half-life, because 0.693 is the natural logarithm of 2.

ROUND

$$k_a = 0.462\ \text{hr}^{-1}$$

Her worked value is 0.462 per hour, to three decimal places, the same precision she uses for rate constants elsewhere in this lecture.

> **DOES THE NUMBER MAKE SENSE?**
> 90 minutes is 1.5 hr, so k_a is 0.693 over 1.5, 0.462 hr^{-1}; a half-life over an hour means k_a below 0.693, and 1.5 × 0.462 returns 0.693.

Problem 1 Worked by the drill

Answer: 3.72 hr

WHAT THE STEM GIVES

D_0	500-mg	*not needed: the dose scales the concentration, not the time*
F	88%	*not needed*

t½a	90 minutes	becomes 1.5 hr, so $k_a = \frac{0.693}{1.5} = 0.462\ hr^{-1}$
t½	5 hours	gives $k = \frac{0.693}{5} = 0.1386\ hr^{-1}$
V_D	20 L	*not needed: t_{max} depends on k_a and k only*

THE WORKING, ONE LINE AT A TIME

UNIT

90 min = 1.5 hr; the elimination half-life is already 5 hr

One half-life is given in minutes and the other in hours, and the two rate constants derived from them are about to be subtracted from one another. Converting the 90 minutes to 1.5 hours first puts both on the same time base.

SETUP

$$t_{max} = \frac{\ln(k_a / k)}{k_a - k}$$

The peak occurs where the absorption and elimination rates are equal, and this is the expression obtained from setting the derivative of the concentration equation to zero. The dose, the bioavailability and the volume of distribution all cancel out and are not used in this part.

ALGEBRA

$$k_a = \frac{0.693}{1.5\ hr} = 0.462\ hr^{-1};\ k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

The expression for the time to peak uses rate constants, so both half-lives have to be converted. Each is 0.693 divided by its own half-life, and the shorter half-life produces the larger constant, which is absorption.

ALGEBRA

$$t_{max} = \frac{\ln(0.462 / 0.1386)}{(0.462 - 0.1386)\ hr^{-1}} = \frac{\ln(3.3333)}{0.3234\ hr^{-1}}$$

The ratio of the rate constants is dimensionless and equals 10 divided by 3. Dividing its logarithm by the difference in reciprocal hours leaves hours.

ALGEBRA

$$t_{max} = \frac{1.2040}{0.3234\ hr^{-1}} = 3.7229\ hr$$

The numerator is the natural logarithm of 3.3333, which is 1.2040, and dividing it by 0.3234 reciprocal hours leaves hours. Rounding the logarithm early to 1.20 would give 3.7105 hours, the same figure to one decimal place, so the point at which the rounding is done does not move the reported answer.

ROUND

$t_{max} = 3.72$ hours

Her worked value is 3.7 hours, to one decimal place, and 3.72 is the same number carried one place further. This peak is much later than the one-hour peak of the earlier antibiotic because absorption here is only about three times faster than elimination rather than twelve.

DOES THE NUMBER MAKE SENSE?

> k_a = 0.462 and k = 0.1386 hr^{-1}, a ratio of 3.3333; ln(3.3333) = 1.2040 over 0.3234 is about 3.72 hr. The peak comes after the 1.5-hr absorption half-life and before the 5-hour elimination half-life.

Problem 1 Worked by the drill

Answer: 13.13 mg/L

WHAT THE STEM GIVES

D_0	500-mg	in the numerator: 0.88 × 500 × 0.462 = 203.28
F	88%	as 0.88 in the numerator of the prefactor
t½a	90 minutes	becomes 1.5 hr, so $k_a = \frac{0.693}{1.5} = 0.462\ hr^{-1}$
t½	5 hours	gives $k = \frac{0.693}{5} = 0.1386\ hr^{-1}$; $k_a - k = 0.3234$
V_D	20 L	in the denominator of the prefactor: 20 × 0.3234 = 6.468

THE WORKING, ONE LINE AT A TIME

UNIT

$$90\ min = 1.5\ hr,\ so\ k_a = \frac{0.693}{1.5\ hr} = 0.462\ hr^{-1};\ k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

Both rate constants must be in reciprocal hours before the exponentials can be evaluated at a time in hours. The absorption half-life is the one given in minutes.

SETUP

$$C_{max} = \frac{F \cdot k_a \cdot D_0}{V_D(k_a - k)}(e^{-k \cdot t} - e^{-k_a \cdot t})$$ evaluated at $t = t_{max}$, so the time to peak is needed before the concentration can be found

The peak concentration is the full oral equation at the one time the curve turns over, so nothing can be substituted until that time is known. The prefactor holds the dose, F and V_D; the bracket holds the competition between absorption and elimination, and only the bracket depends on time.

ALGEBRA

$$t_{max} = \frac{\ln(0.462 / 0.1386)}{(0.462 - 0.1386)\ hr^{-1}} = 3.7\ hr$$

The peak concentration is the concentration at the time of the peak, so that time is found first and then substituted. She uses 3.7 hours in the substitution that follows.

ALGEBRA

$$Prefactor = \frac{(0.88)(500\ mg)(0.462\ hr^{-1})}{[(20\ L)(0.3234\ hr^{-1})]} = \frac{203.28}{6.468} = 31.429\ mg/L$$

The bioavailable dose is 0.88 of 500 mg, and dividing by the volume of distribution is what turns an amount into a concentration. Milligrams times reciprocal hours over litres times reciprocal hours leaves milligrams per litre.

ALGEBRA

Bracket = $e^{-(0.1386)(3.7)} - e^{-(0.462)(3.7)} = 0.59879 - 0.18096 = 0.41783$

Each exponent is dimensionless. By the time of the peak the absorption term has fallen to about 18 per cent of its starting value while the elimination term is still near 60 per cent, which is why the curve is turning over at this moment.

ALGEBRA

$C_{max} = (31.429 \text{ mg/L})(0.41783) = 13.132 \text{ mg/L}$

Multiplying the prefactor by the dimensionless bracket leaves a concentration in milligrams per litre. Evaluating at the unrounded time to peak of 3.7229 hours instead of 3.7 gives the same figure, so the earlier rounding of t_{max} costs nothing here.

ROUND

$C_{max} = 13.13 \text{ mg/L}$

Report 13.13 mg/L, or about 13 mg/L; two decimal places is the precision the parameters support. The peak sits well below the 22 mg/L the dose and volume would give on instant absorption, because elimination is already removing drug while absorption is still going on.

> **DOES THE NUMBER MAKE SENSE?**
> The prefactor is 31.429 mg/L and the bracket at the peak is 0.41783, so C_{max} is a little over two fifths of 31.429, about 13.13 mg/L; it must be below 0.88 × 500 over 20 = 22 mg/L.

The printed practice sheet gives 40 L; in class she changed it to 20 L, which this question uses. With 40 L the peak is 6.57 mg/L, the next question.

Problem 1 Worked by the drill

Answer: 6.57 mg/L

WHAT THE STEM GIVES

D_0	500-mg	in the numerator of the prefactor, unchanged
V_D	40 L	the new denominator of the prefactor: 40 × 0.3234 = 12.936
V_D before	20 L	*not needed for the number: the new volume is double it*

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{max} = \dfrac{F \cdot k_a \cdot D_0}{V_D(k_a - k)}(e^{-k \cdot t} - e^{-k_a \cdot t})$ at $t = t_{max}$, and only the volume of distribution has changed, so t_{max} is unaffected

The time to peak depends on the two rate constants alone, so doubling the volume of distribution moves the height of the curve without moving the moment it peaks. The volume sits in the prefactor only, which is why the bracket can be carried over unchanged from the previous part rather than recomputed.

ALGEBRA

$$t_{max} = \frac{\ln(0.462 / 0.1386)}{(0.462 - 0.1386)\ hr^{-1}} = 3.7\ hr, \text{ unchanged}$$

The volume of distribution does not appear in the expression for the time to peak, so doubling it cannot move the peak in time. Only the height of the curve is affected, which is why the same time is substituted as before.

ALGEBRA

$$\text{Prefactor} = \frac{(0.88)(500\ mg)(0.462\ hr^{-1})}{[(40\ L)(0.3234\ hr^{-1})]} = \frac{203.28}{12.936} = 15.714\ mg/L$$

Doubling the volume in the denominator halves the prefactor, because the same bioavailable amount of drug is now spread through twice the apparent volume. The units are unchanged.

ALGEBRA

$$\text{Bracket} = e^{-(0.1386)(3.7)} - e^{-(0.462)(3.7)} = 0.41783$$

The bracket contains only the rate constants and the time, none of which has changed, so it takes exactly the same value as with the smaller volume.

ALGEBRA

$$C_{max} = (15.714\ mg/L)(0.41783) = 6.566\ mg/L$$

The bracket is unchanged from the 20 L version because it depends only on the rate constants and the time, so only the prefactor has halved. Multiplying a concentration by a dimensionless fraction leaves a concentration.

ROUND

$$C_{max} = 6.57\ mg/L$$

This is exactly half the 13.13 mg/L obtained with a 20 L volume, as it must be when the only change is a doubling of the volume in the denominator. Doubling the volume of distribution halves every concentration on the curve without moving the time at which the peak occurs.

> **DOES THE NUMBER MAKE SENSE?**
> Doubling the volume halves the prefactor and leaves t_{max} and the bracket unchanged, so the peak halves: 15.714 × 0.41783 is about 6.57 mg/L, below 0.88 × 500 over 40 = 11 mg/L.

This is the printed practice sheet's version, with a volume of distribution of 40 L. In class she used 20 L, which gives 13.13 mg/L.

Problem 2 Worked by the drill

Answer: 3.81 hr

WHAT THE STEM GIVES

D_0	750-mg	*not needed: the half-life comes from an exponent*
coefficient	23.2	*not needed*
k	0.182	the smaller exponent, in hr^{-1}, the denominator of $\frac{0.693}{k}$
k_a	0.872	*not needed: the larger exponent is absorption*

THE WORKING, ONE LINE AT A TIME

SETUP

$k = 0.182\ hr^{-1}$, the smaller of the two rate constants

Elimination is the slower process for an ordinary oral product, so the smaller exponential constant is k. Using the larger one would give the absorption half-life instead, which is less than one hour here.

ALGEBRA

$$t½ = \frac{0.693}{0.182\ hr^{-1}}$$

First-order half-life is 0.693 divided by the rate constant, and this relation is not supplied on the equation sheet. A pure number over reciprocal hours leaves hours.

ROUND

$t½ = 3.8077$ hr, reported as 3.8 hours

Report 3.81 hours, or 3.8 to one decimal place. The unrounded division, 0.693 over 0.182 per hour, gives 3.8077 hours.

DOES THE NUMBER MAKE SENSE?

The smaller exponent, $0.182\ hr^{-1}$, is k; 0.693 over 0.182 is a little under 4 hr, 3.8 hours, longer than the absorption half-life from 0.872.

Problem 2

Worked by the drill

Answer: 2.27 hr

WHAT THE STEM GIVES

D_0	750-mg	*not needed: t_{max} depends on k_a and k only*
coefficient	23.2	*not needed*
k	0.182	the smaller exponent, in hr^{-1}: the denominator of the ratio, subtracted below
k_a	0.872	the larger exponent, in hr^{-1}: the numerator of the ratio; $k_a - k = 0.690$

THE WORKING, ONE LINE AT A TIME

SETUP

$k_a = 0.872\ hr^{-1}$ and $k = 0.182\ hr^{-1}$

Both constants are read from the exponents, the larger being absorption. No unit conversion is needed because the stem states that time is in hours.

SETUP

$$t_{max} = \frac{\ln(k_a / k)}{k_a - k}$$

The time to peak comes from the two rate constants alone. The leading coefficient of 23.2 plays no part, which is why this step can be done before anything is known about the dose or the volume of distribution.

ALGEBRA

$$t_{max} = \frac{\ln(0.872 / 0.182)}{(0.872 - 0.182)\ hr^{-1}} = \frac{\ln(4.7912)}{0.690\ hr^{-1}}$$

The ratio is dimensionless and its logarithm is 1.5668. Dividing by a difference in reciprocal hours returns a time in hours.

ALGEBRA

$$t_{max} = \frac{1.5668}{0.690\ hr^{-1}} = 2.2707\ hr$$

The ratio of the two rate constants is 4.7912 and its natural logarithm is 1.5668, divided by their difference of 0.690 reciprocal hours. Reciprocal hours in the denominator invert to give hours, which is what a time to peak must carry.

ROUND

$t_{max} = 2.27$ hours

Her worked value is 2.27 hours, to two decimal places. This peak is later than the one-hour peak of the earlier antibiotic because here the two rate constants are much closer together, so absorption and elimination stay in competition for longer.

> **DOES THE NUMBER MAKE SENSE?**
> k_a over k is 4.7912, so ln(4.7912) = 1.5668 over 0.690 is about 2.27 hr; the peak must come after the absorption half-life, 0.693 over 0.872, under an hour.

Problem 2

Worked by the drill

Answer: 12.14 mg/L

WHAT THE STEM GIVES

D_0	750-mg	*not needed: the printed equation already carries the dose*
coefficient	23.2	in mg/L, multiplied by the bracket at t_{max}
k	0.182	the smaller exponent, in hr^{-1}: gives t_{max} and $e^{-0.41314}$
k_a	0.872	the larger exponent, in hr^{-1}: gives t_{max} and $e^{-1.97944}$

THE WORKING, ONE LINE AT A TIME

SETUP

C_{max} is the concentration at t_{max}, so the time to peak has to be found first and then substituted back into the equation the stem supplies

The peak is not a separate parameter to look up; it is the concentration equation evaluated at the one time the curve turns over. So the problem runs in two stages, and the second cannot start until the first gives a number. Naming both stages first stops the two rate constants being swapped.

ALGEBRA

$$t_{max} = \frac{\ln(0.872 / 0.182)}{(0.872 - 0.182)\ hr^{-1}} = 2.27\ hr$$

The peak concentration is the concentration at the time of the peak, so that time has to be found before the substitution. She calculates it first and then reuses it in both exponentials.

SETUP

$C_{max} = 23.2(e^{-(0.182)(2.27)} - e^{-(0.872)(2.27)})$

The prefactor is already collected into the 23.2, so the time to peak is substituted into both exponentials and nothing else is needed. Each exponent is a reciprocal time multiplied by a time and is dimensionless.

ALGEBRA

$e^{-0.41314} = 0.66157$; $e^{-1.97944} = 0.13815$; difference = 0.52342

At the peak the elimination term is still at about two thirds of its starting value while the absorption term has fallen to about one seventh, and the gap between them is what the concentration is proportional to.

ALGEBRA

$C_{max} = (23.2)(0.52342) = 12.143$ mg/L

The intercept of 23.2 multiplies a dimensionless bracket, so the result keeps the concentration units of the equation. Evaluating at the unrounded time to peak gives the same figure, so the rounding of t_{max} to two decimals costs nothing.

ROUND

$C_{max} = 12.14$ mg/L

Her worked value is 12.14 milligrams per litre, which she also states as micrograms per millilitre, the two being numerically identical. Two decimal places matches the precision of the coefficients printed in the equation she supplies.

> **DOES THE NUMBER MAKE SENSE?**
> At $t_{max} = 2.27$ hr, $e^{-0.41314}$ is 0.66157 and the absorption term 0.13815 is still present, so C_{max} is a little over half of 23.2 mg/L; it must be below the coefficient 23.2.

Problem 2 Worked by the drill

Answer: 34.32 L

WHAT THE STEM GIVES

D_0	750-mg	in the numerator: 0.84 × 750 × 0.872 = 549.36
F	84%	as 0.84 in the numerator
coefficient	23.2	in mg/L, set equal to the prefactor and moved to the denominator
k	0.182	the smaller exponent, in hr^{-1}: $k_a - k = 0.690$
k_a	0.872	the larger exponent, in hr^{-1}, in the numerator and in $k_a - k$

THE WORKING, ONE LINE AT A TIME

SETUP

$k_a = 0.872\ hr^{-1}$ and $k = 0.182\ hr^{-1}$, identified by size

The larger of the two exponential constants is the absorption rate constant for an ordinary oral product, and the prefactor expression needs k_a in its numerator. Assigning them the other way round would put 0.182 on top and make the difference negative, giving a negative volume.

SETUP

$$23.2\ \text{mg/L} = \frac{F \cdot k_a \cdot D_0}{(V_D(k_a - k))}, \text{ so } V_D = \frac{F \cdot k_a \cdot D_0}{[23.2\ \text{mg/L} \times (k_a - k)]}$$

The leading coefficient of the equation is the whole constant prefactor, so equating the two and rearranging isolates the volume of distribution. Everything else in the expression is given in the stem.

ALGEBRA

$$V_D = \frac{(0.84)(750\ \text{mg})(0.872\ \text{hr}^{-1})}{[(23.2\ \text{mg/L})(0.690\ \text{hr}^{-1})]} = \frac{549.36}{16.008}$$

The difference of the rate constants is 0.872 minus 0.182, which is 0.690 per hour. Milligrams times reciprocal hours divided by milligrams per litre times reciprocal hours leaves litres.

ROUND

$V_D = 34.32$ L, reported as 34.3 L

Report 34.32 litres, or 34.3 to one decimal place. The unrounded division gives 34.3178 litres, and milligrams divided by milligrams per litre leaves litres.

DOES THE NUMBER MAKE SENSE?

The coefficient 23.2 mg/L equals $F \cdot k_a \cdot D_0$ over $V_D(k_a - k)$. k_a over $(k_a - k)$, 0.872 over 0.690, is a little over 1, so V_D is a little over 0.84 × 750 over 23.2 mg/L; 34.32 L fits.